

Duo Xu

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RESEARCH INTERESTS

Vision-guided robotics; language-conditioned manipulation; visual feedback, calibration, and safe physical control; physics-guided computational imaging; automatic optical alignment; image-quality assessment; machine vision; learning-assisted modeling and optimization for real optical and robotic systems.

EDUCATION

**Tianjin University | M.Eng. in Electronic Information,
recommended admission**

Sep 2024 - Present

Advisor: Prof. Xiaodong Zhang

- Average score: 87.88/100; rank: 15/115.
- Research focus: freeform ultra-short-throw projection systems, image-quality-driven alignment, optical simulation, and closed-loop optimization.

**University of Science and Technology Beijing | B.Eng. in
Mechanical Engineering**

Sep 2020 - Jul 2024

- Average score: 83.62/100; rank: 17/178.
- Completed second-major coursework in Computer Science and Technology, including programming, data structures, image processing, and computational engineering methods.

PUBLICATIONS AND INTELLECTUAL PROPERTY

- Jiang Li*, Duo Xu*, Changshuai Fang, Zexiao Li, and Xiaodong Zhang. "Sensitivity-informed multi-objective adaptive weighting method for automatic alignment of freeform USTP systems." *Optics Express* 34(11), 20358-20378, 2026. DOI: 10.1364/OE.34.020358. *Equal contribution.
- Patent application under review: "An optimization and alignment method for imaging and display systems based on physical-model-driven adaptive weighting" (translated title).

TECHNICAL SKILLS

- Programming and engineering software: C++, Qt, Python, MATLAB, CMake, Git/GitLab; closed-loop software development, hardware communication, automation scripts, and technical documentation.
- Computer vision and optical tools: OpenCV, Halcon, Zemax OpticStudio, ZOS-API, FDTD Lumerical; structured-target analysis, geometric distortion, MTF-related image-quality assessment, and 6-DOF error analysis.
- Robotics and embodied AI: LeRobot, SO-101 leader-follower teleoperation, PyTorch, Hugging Face, real-robot data collection, ACT, SmoI/VLA, and multimodal rollout evaluation.

RESEARCH AND PROJECT EXPERIENCE

**Automatic Alignment and Image-quality Optimization of
Freeform USTP Systems**

2024 - Present

Tianjin University | Master's research project

- Built a Zemax non-sequential optical model matched to a physical prototype and simulated 6-DOF freeform-mirror pose perturbations.
- Designed structured-target image acquisition and analysis to evaluate geometric distortion and MTF-related image quality; implemented a C++ multi-objective adaptive weighting optimizer for closed-loop alignment.
- Integrated optical simulation, industrial-camera acquisition, a 6-DOF precision platform, image-quality evaluation, optimization software, and motion execution. In 10 repeated experiments, final mean MTF exceeded 70% and each alignment run took about 10 minutes.

**Real-Robot Language-Conditioned Manipulation and VLA
Failure Analysis**

Jul 2026 - Present

SO-101 / LeRobot research platform

- Built and calibrated a dual-view SO-101 leader-follower manipulation platform with external and wrist RGB cameras; implemented teleoperation, synchronized RGB/state/action recording, and hardware-safe real-robot rollout workflows.
- Curated a 60-episode dual-camera dataset for three language-conditioned color-sorting instructions, with episode-level train/validation/test splits and reproducible Hugging Face dataset/model releases.
- Established a visual imitation-learning ACT baseline with 8/10 real-robot success on a fixed pick-and-place task; fine-tuned and deployed SmoI/VLA for multi-instruction manipulation and structured rollout failure analysis.

Wavefront-aberration-guided Automatic Alignment of Optical Lens Groups

2024 - Present

Tianjin University | Optical alignment software and experiment

- Modeled the relationship between low-order Zernike coefficients and lens-group misalignment parameters; developed Qt/C++ modules using ZOS-API for perturbation scanning, ray tracing, image analysis, and result export.
- Supported phase-deflectometry experiments for off-axis optical components, including acquisition-workflow debugging and simulation-experiment comparison.

Chromatic-aberration-enhanced Metalens Design and Simulation for Spectral Confocal Sensing

Mar 2024 - Oct 2024

Tianjin University and USTB | Undergraduate thesis project

- Designed a metasurface-based spectral confocal sensing solution; derived wavelength-dependent phase profiles and used FDTD Lumerical and MATLAB to evaluate wavelength separation, focal positions, and optical fields.

FPGA-based Line-scan Vision Inspection System

Sep 2024 - Oct 2024

Tianjin University | Collaborative research project with Hexagon

- Developed C++ modules for high-speed line-scan image reconstruction, preprocessing, feature extraction, candidate-region analysis, and result output; adapted selected algorithms into HLS/FPGA-friendly streaming blocks.

RESEARCH STRENGTHS AND PREPARATION

- Vision-feedback closed-loop system building: connects camera acquisition, quantitative image analysis, optimization, and physical motion execution.
- Experimental discipline for physical systems: calibration, repeatable task layouts, episode-level evaluation, failure analysis, and explicit limits on unverified autonomy.

ENGLISH PROFICIENCY AND AWARDS

- IELTS Academic 6.5 (Listening 7.5, Reading 7.0, Writing 6.5, Speaking 5.5); CET-6 and CET-4 passed.
- Second-Class Academic Scholarship, Tianjin University (2025); People's Scholarship, USTB (2021-2023); Third Prize, National College Students Mathematics Competition (2021).